Лабораторная работа №5

Моделирование сетей передачи данных

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Цель работы

Основной целью работы является получение навыков проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных. Эти параметры влияют на производительность протоколов и сетей.

Задание

1. Задайте простейшую топологию, состоящую из двух хостов и коммутатора с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.
3. Реализуйте воспроизводимый эксперимент по добавлению правила отбрасывания пакетов в эмулируемой глобальной сети.
4. Самостоятельно реализуйте воспроизводимые эксперименты по исследованию параметров сети, связанных с потерей, изменением порядка и повреждением пакетов при передаче данных.

Информацию о сетевых интерфейсах и IP-адресах хостов

"host: h1"@mininet-vm

```
root@mininet-vm:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.1  netmask 255.0.0.0  broadcast 10.255.255.255
        ether b2:ba:d4:f7:06:a2  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 918  bytes 261472 (261.4 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 918  bytes 261472 (261.4 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet#
```

"host: h2"@mininet-vm

```
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.2  netmask 255.0.0.0  broadcast 10.255.255.255
        ether 3e:89:46:63:84:ad  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 919  bytes 262532 (262.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 919  bytes 262532 (262.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet#
```

Проверка соединения между хостами

"host: h1"@mininet-vm

```
       TX packets 0  bytes 0 (0.0 B)
       TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
       inet 127.0.0.1  netmask 255.0.0.0
       loop  txqueuelen 1000  (Local Loopback)
       RX packets 918  bytes 261472 (261.4 KB)
       RX errors 0  dropped 0  overruns 0  frame 0
       TX packets 918  bytes 261472 (261.4 KB)
       TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 6
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=8.16 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.520 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.072 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.244 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.829 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.081 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5065ms
rtt min/avg/max/mdev = 0.072/1.651/8.162/2.923 ms
```

"host: h2"@mininet-vm

```
       TX packets 0  bytes 0 (0.0 B)
       TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
       inet 127.0.0.1  netmask 255.0.0.0
       loop  txqueuelen 1000  (Local Loopback)
       RX packets 919  bytes 262532 (262.5 KB)
       RX errors 0  dropped 0  overruns 0  frame 0
       TX packets 919  bytes 262532 (262.5 KB)
       TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.79 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.094 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.082 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.081 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.090 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.094 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5089ms
rtt min/avg/max/mdev = 0.081/0.538/2.789/1.006 ms
root@mininet-vm:/home/mininet#
```

Добавление потери пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 10
%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 100
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=6.17 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.948 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.358 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.083 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.159 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.124 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.110 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.079 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.093 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.111 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.201 ms
```

Рисунок 3: Добавление потери пакетов

Добавление потери пакетов

```
64 bytes from 10.0.0.2: icmp_seq=89 ttl=64 time=0.144 ms
64 bytes from 10.0.0.2: icmp_seq=90 ttl=64 time=0.114 ms
64 bytes from 10.0.0.2: icmp_seq=91 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=92 ttl=64 time=0.083 ms
64 bytes from 10.0.0.2: icmp_seq=93 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=94 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=95 ttl=64 time=0.098 ms
64 bytes from 10.0.0.2: icmp_seq=96 ttl=64 time=0.145 ms
64 bytes from 10.0.0.2: icmp_seq=97 ttl=64 time=0.190 ms
64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.108 ms
64 bytes from 10.0.0.2: icmp_seq=100 ttl=64 time=0.090 ms

--- 10.0.0.2 ping statistics ---
100 packets transmitted, 87 received, 13% packet loss, time 101326ms
rtt min/avg/max/mdev = 0.069/0.199/6.173/0.651 ms
root@mininet-vm:/home/mininet#
```

Рисунок 4: Результат добавления потери пакетов

Добавление потери пакетов

```
64 bytes from 10.0.0.2: icmp_seq=94 ttl=64 time=0.096 ms
64 bytes from 10.0.0.2: icmp_seq=95 ttl=64 time=0.079 ms
64 bytes from 10.0.0.2: icmp_seq=97 ttl=64 time=0.138 ms
64 bytes from 10.0.0.2: icmp_seq=98 ttl=64 time=0.090 ms
64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=100 ttl=64 time=0.117 ms

--- 10.0.0.2 ping statistics ---
100 packets transmitted, 86 received, 14% packet loss, time 1013
rtt min/avg/max/mdev = 0.073/0.183/5.740/0.611 ms
root@mininet-vm:/home/mininet#
```

```
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 919  bytes 262532 (262.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 919  bytes 262532 (262.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collision

root@mininet-vm:/home/mininet# ping 10.0.0.1 -c 6
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.79 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.094 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.082 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.081 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.090 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.094 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5089ms
rtt min/avg/max/mdev = 0.081/0.538/2.789/1.006 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 ro
%
root@mininet-vm:/home/mininet#
```

Удаление правил

```
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=6.02 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.54 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.293 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.099 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.099 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.072 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.176 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.083 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.208 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.097 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.258 ms
^C
--- 10.0.0.2 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11203ms
rtt min/avg/max/mdev = 0.072/0.752/6.016/1.634 ms
root@mininet-vm:/home/mininet#
```

Рисунок 6: Удаление правил

Добавление значения корреляции для потери пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 50
% 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 50
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.307 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.095 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.091 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.119 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.220 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.092 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.105 ms
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.119 ms
64 bytes from 10.0.0.2: icmp_seq=26 ttl=64 time=0.137 ms
```

Рисунок 7: Добавление значения корреляции для потери пакетов

Добавление значения корреляции для потери пакетов

```
"host: h1"@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.307 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.095 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.091 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.119 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.220 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.092 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.105 ms
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.119 ms
64 bytes from 10.0.0.2: icmp_seq=26 ttl=64 time=0.137 ms
64 bytes from 10.0.0.2: icmp_seq=27 ttl=64 time=0.217 ms
64 bytes from 10.0.0.2: icmp_seq=28 ttl=64 time=0.100 ms
64 bytes from 10.0.0.2: icmp_seq=31 ttl=64 time=0.132 ms
64 bytes from 10.0.0.2: icmp_seq=35 ttl=64 time=0.106 ms
64 bytes from 10.0.0.2: icmp_seq=37 ttl=64 time=0.109 ms
64 bytes from 10.0.0.2: icmp_seq=39 ttl=64 time=0.123 ms
64 bytes from 10.0.0.2: icmp_seq=40 ttl=64 time=0.190 ms
64 bytes from 10.0.0.2: icmp_seq=41 ttl=64 time=0.094 ms
64 bytes from 10.0.0.2: icmp_seq=50 ttl=64 time=0.105 ms

--- 10.0.0.2 ping statistics ---
50 packets transmitted, 19 received, 62% packet loss, time 50150ms
rtt min/avg/max/mdev = 0.085/0.134/0.307/0.056 ms
root@mininet-vm:/home/mininet#
```

Рисунок 8: Результат при корреляции

Добавление повреждения пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem corrupt
 0.01%
```

Рисунок 9: Добавление повреждения пакетов

Добавление повреждения пакетов

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 34630 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.00   sec   876 MBytes  7.35 Gbits/sec    3    799 KBytes
[  7]   1.00-2.00   sec   913 MBytes  7.66 Gbits/sec    3    452 KBytes
[  7]   2.00-3.00   sec   967 MBytes  8.10 Gbits/sec    0    990 KBytes
[  7]   3.00-4.00   sec   890 MBytes  7.46 Gbits/sec    1    991 KBytes
[  7]   4.00-5.01   sec   954 MBytes  7.93 Gbits/sec    1    996 KBytes
[  7]   5.01-6.00   sec   891 MBytes  7.56 Gbits/sec    1    799 KBytes
[  7]   6.00-7.00   sec   906 MBytes  7.60 Gbits/sec    2    786 KBytes
[  7]   7.00-8.00   sec   895 MBytes  7.50 Gbits/sec    2    537 KBytes
[  7]   8.00-9.00   sec   839 MBytes  7.04 Gbits/sec    0    822 KBytes
[  7]   9.00-10.00  sec   775 MBytes  6.50 Gbits/sec    2    718 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-10.00  sec  8.70 GBytes  7.47 Gbits/sec   15             sender
[  7]   0.00-10.01  sec  8.70 GBytes  7.46 Gbits/sec                  receiver

iperf Done.
root@mininet-vm:/home/mininet# 
```

```
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 10.0.0.1, port 34628
[  7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 34630
[ ID] Interval           Transfer     Bitrate
[  7]   0.00-1.00   sec   874 MBytes  7.33 Gbits/sec
[  7]   1.00-2.00   sec   912 MBytes  7.65 Gbits/sec
[  7]   2.00-3.00   sec   965 MBytes  8.10 Gbits/sec
[  7]   3.00-4.00   sec   890 MBytes  7.47 Gbits/sec
[  7]   4.00-5.00   sec   951 MBytes  7.98 Gbits/sec
[  7]   5.00-6.00   sec   896 MBytes  7.52 Gbits/sec
[  7]   6.00-7.00   sec   904 MBytes  7.58 Gbits/sec
[  7]   7.00-8.00   sec   897 MBytes  7.52 Gbits/sec
[  7]   8.00-9.00   sec   841 MBytes  7.05 Gbits/sec
[  7]   9.00-10.00  sec   771 MBytes  6.47 Gbits/sec
[  7]  10.00-10.01  sec  5.56 MBytes  5.13 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  7]   0.00-10.01  sec  8.70 GBytes  7.46 Gbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Добавление переупорядочивания пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 1
0ms reorder 25% 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=11.6 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=12.0 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.246 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=11.3 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=10.9 ms
```

Рисунок 11: Добавление переупорядочивания пакетов

Добавление переупорядочивания пакетов

```
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.246 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=11.3 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=10.5 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=10.5 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=11.3 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=12.4 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=11.3 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=10.9 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19083ms
rtt min/avg/max/mdev = 0.085/9.932/12.351/3.291 ms
root@mininet-vm:/home/mininet#
```

Рисунок 12: Результат добавления переупорядочивания пакетов

Добавление дублирования пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem duplica
te 50%
root@mininet-vm:/home/mininet# ping 10.0.0.2 -c 20
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.432 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.093 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.094 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.091 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.315 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.105 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.527 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.528 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.139 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.103 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.104 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.129 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.131 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.316 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.317 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.101 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.079 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.113 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.077 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.078 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.241 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.241 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.090 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.091 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.142 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.103 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.103 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.244 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, +10 duplicates, 0% packet loss, time 19429m
s
rtt min/avg/max/mdev = 0.077/0.176/0.528/0.130 ms
root@mininet-vm:/home/mininet#
```

Скрипт

mininet@mininet-vm: ~/work/lab_netem_ii/simple-drop

```
  GNU nano 4.8                    lab_netem_ii.py                    Modified
#!/usr/bin/env python

"""
Simple experiment.
Output: ping.dat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
import time

def emptyNet():

        "Create an empty network and add nodes to it."

        net = Mininet( controller=Controller, waitConnected=True )

        info( '*** Adding controller\n' )
        net.addController( 'c0' )

        info( '*** Adding hosts\n' )
        h1 = net.addHost( 'h1', ip='10.0.0.1' )
        h2 = net.addHost( 'h2', ip='10.0.0.2' )

        info( '*** Adding switch\n' )
        s1 = net.addSwitch( 's1' )

        info( '*** Creating links\n' )
        net.addLink( h1, s1 )
        net.addLink( h2, s1 )

        info( '*** Starting network\n')
        net.start()

        info( '*** Set delay\n')
        h1.cmdPrint( 'tc qdisc add dev h1-eth0 root netem loss 10%' )
        h2.cmdPrint( 'tc qdisc add dev h2-eth0 root netem loss 10%' )

        time.sleep(10)

        info( '*** Ping\n')
        h1.cmdPrint( 'ping -c 100', h2.IP(), '| grep "time=" | awk \'{print $5, $7}\' >

        info( '*** Stopping network' )
        net.stop()

if __name__ == '__main__':
        setLogLevel( 'info' )
        emptyNet()
```

Проведение эксперимента

```
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ nano lab_netem_ii.py
mininet@mininet-vm:~/work/lab_netem_ii/simple-drop$ make
sudo python lab_netem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem loss 10%',)
*** h2 : ('tc qdisc add dev h2-eth0 root netem loss 10%',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "packet loss" | awk \'{print $6, $7, $8}\'
 > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
```

Рисунок 15: Проведение эксперимента

Проведение эксперимента

```
mininet@mininet-vm:~/work/lab_netem_ii/correlation-drop$ nano lab_netem_ii.py
mininet@mininet-vm:~/work/lab_netem_ii/correlation-drop$ make
sudo python lab_netem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem loss 50% 50%',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "packet loss" | awk \'{print $6, $7, $8}\'
 > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
```

Рисунок 16: Проведение эксперимента

Проведение эксперимента

```
mininet@mininet-vm:~/work/lab_netem_ii/reorder-drop$ make
sudo python lab_netem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem delay 10ms reorder 25% 50%',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "packet loss" | awk \'{print $6, $7,
 > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
```

Рисунок 17: Проведение эксперимента

Проведение эксперимента

```
mininet@mininet-vm:~/work/lab_netem_ii/corrupt-drop$ make
sudo python lab_netem_ii.py
*** Adding controller
*** Adding hosts
*** Adding switch
*** Creating links
*** Starting network
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Set delay
*** h1 : ('tc qdisc add dev h1-eth0 root netem corrupt 0.01%',)
*** Ping
*** h1 : ('ping -c 100', '10.0.0.2', '| grep "packet loss" | awk \'{print $6, $7, $8}\'
 > ping.dat')
*** Stopping network*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
sudo chown mininet:mininet ping.dat
```

Рисунок 18: Проведение эксперимента

Выводы

В ходе выполнения лабораторной работы я получила навыки проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.